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Abstract

Zika virus (ZIKV) and dengue virus (DENV) are mosquito-borne flaviviruses and emerging pathogens which have caused severe disease in tropical and subtropical regions. A major complication for ZIKV and DENV vaccine development is the potential for immunogens to enhance subsequent infection by antibody-dependent enhancement (ADE). In prior work, our group used structure-based protein engineering to develop a “resurfaced” (rs) DENV E-glycoprotein domain III (DIII) immunogen where neutralizing epitopes which may be targeted by ADE antibodies were masked. We showed that mice immunized with one DENV2 DIII mutant (rs2) elicit a potent cross-neutralizing antibody response across DENV1-3.

Recently emerging literature has provided evidence for the protective role of T-cells during DENV and ZIKV infections. In this study we hypothesize that the multivalent display of both B-cell and T-cell epitopes as a polypeptide-protein nanoparticle assembly will promote an immunofocused and more expansive adaptive immune response, compared to rs2 nanoparticle alone. We aim to engineer a multivalent nanoparticle which stimulates a potent and broad adaptive immune response across ZIKV and DENV infections. This will provide important insights into the development of a promising pan-vaccine candidate.

Background

- ZIKV has caused epidemics in parts of Asia, Africa and the Americas. Patients diagnosed with ZIKV may experience joint and muscle pains, while severe ZIKV infections may lead to neurological complications such as congenital ZIKV syndrome and Guillain-Barre syndrome.
- DENV affects half of the global population and co-circulates as four serotypes (DENV1 – DENV4). Mild dengue infections are also characterized by body aches, but some DENV infections may cause severe dengue syndrome, resulting in hemorrhagic fever and organ failure.
- The vector of both viruses is the *Aedes* mosquito. As a result of climate change, increased urbanization, travel and mosquito resistance to insecticides, these vectors have spread to new regions, increasing the prevalence of ZIKV and DENV infections.
- ADE complicates ZIKV/DENV vaccine development. It is a phenomenon caused by cross-reactive and poorly neutralizing antibodies elicited during a primary infection, which promote increased viral pathogenesis during a secondary infection by a heterologous virus.

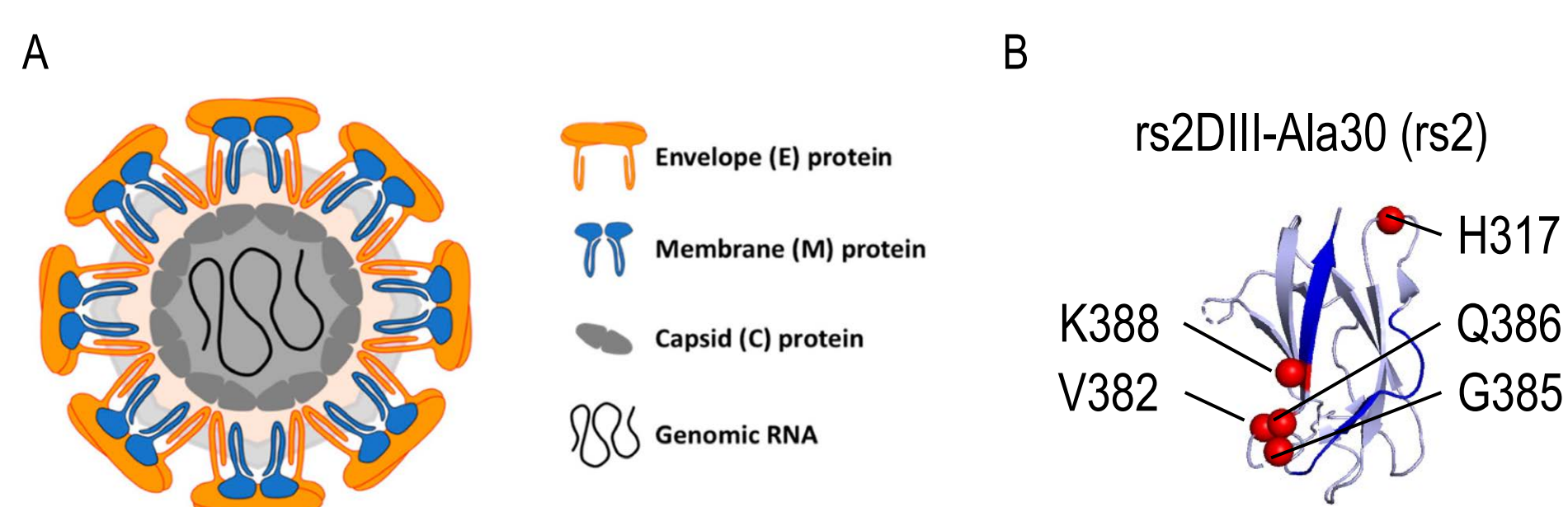


Figure 1. ZIKV and E DIII structures (A) ZIKV virion structure. ¹ (B) Resurfaced (rs) DENV2 DIII Ala30 (rs2).

- Using structure-based protein design, we developed a resurfaced DENV2 DIII mutant (rs2) where the A/G strand epitope targeted by neutralizing antibodies was maintained, while serotype-specific and non-neutralizing epitopes were masked (Fig 1).

Background cont'd

- Multivalent display of rs2 on Lumazine Synthase (LS) nanoparticles elicits a potent cross-neutralizing antibody response across DENV1-3.
- Preliminary studies by Shresta et. al. on ZIKV and DENV T-cell immune response have demonstrated the importance of T-cell epitopes in non-structural proteins of ZIKV that can elicit ZIKV-DENV cross-reactive CD8+ T-cell activity ².

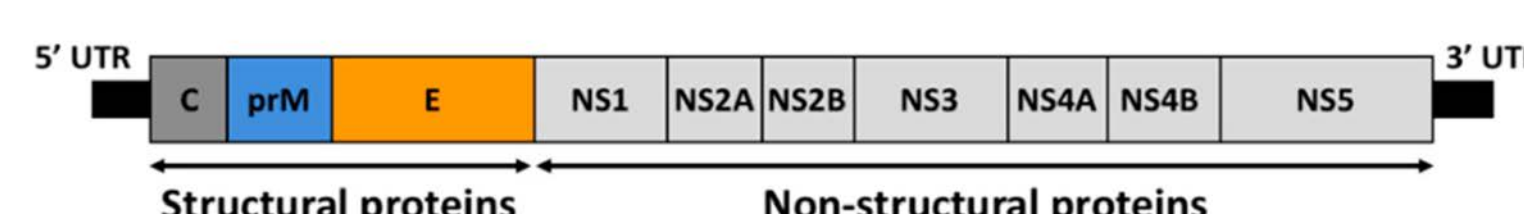


Figure 2. ZIKV genome organization.¹ Target B-cell epitopes are located in the E structural protein region while target T-cell epitopes are located in the non-structural protein 5 (NS5) region.

- Leading T-cell polypeptides of interest:
 - NS5 2793-2792 [CAEAPNMKVI]** - CD8+ eliciting ZIKV Non-structural Protein 5 (NS5) region (Fig 2).
 - PADRE [AKFVAAWTLKAAA]** - Pan DR-binding epitope is a universal synthetic peptide which activates antigen specific CD4+ T-cells.

Results & Discussion

- LS nanoparticles were expressed with T-cell polypeptides (Fig 3 & 4).

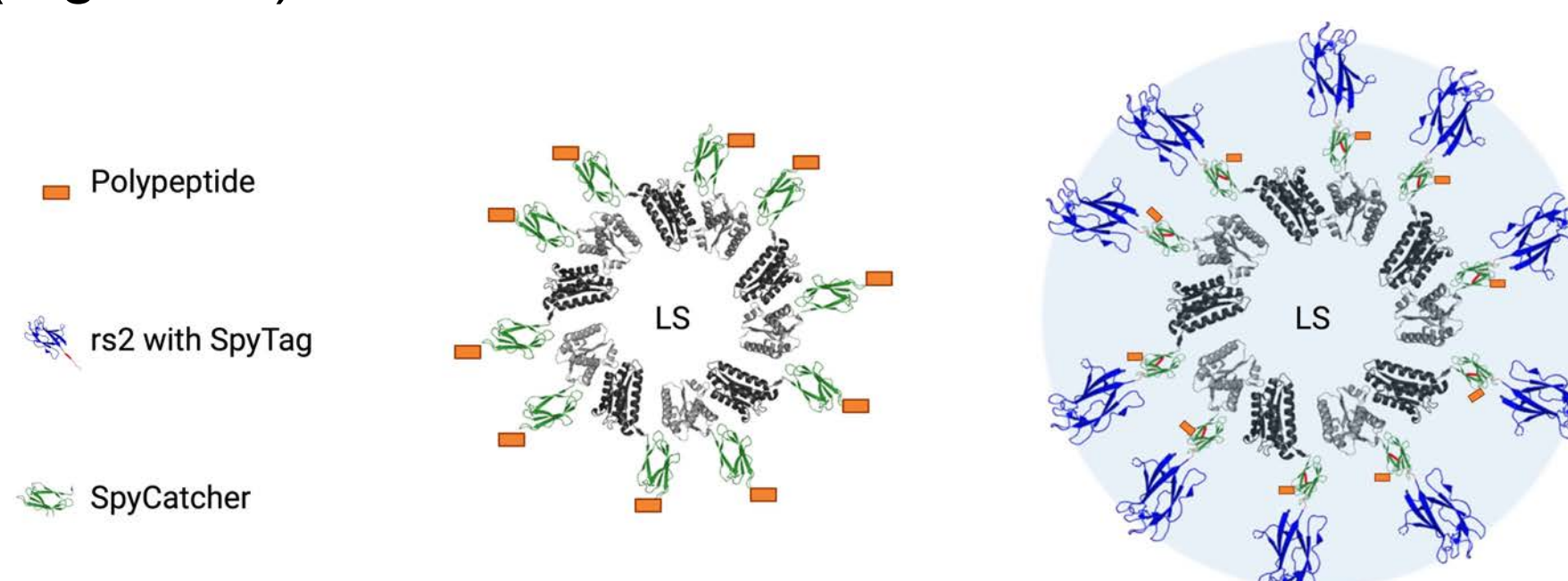


Figure 3. LS nanoparticle with T-cell polypeptides.³ (A) Unconjugated LS-polypeptide nanoparticle. (B) LS-polypeptide nanoparticle with conjugated rs2.

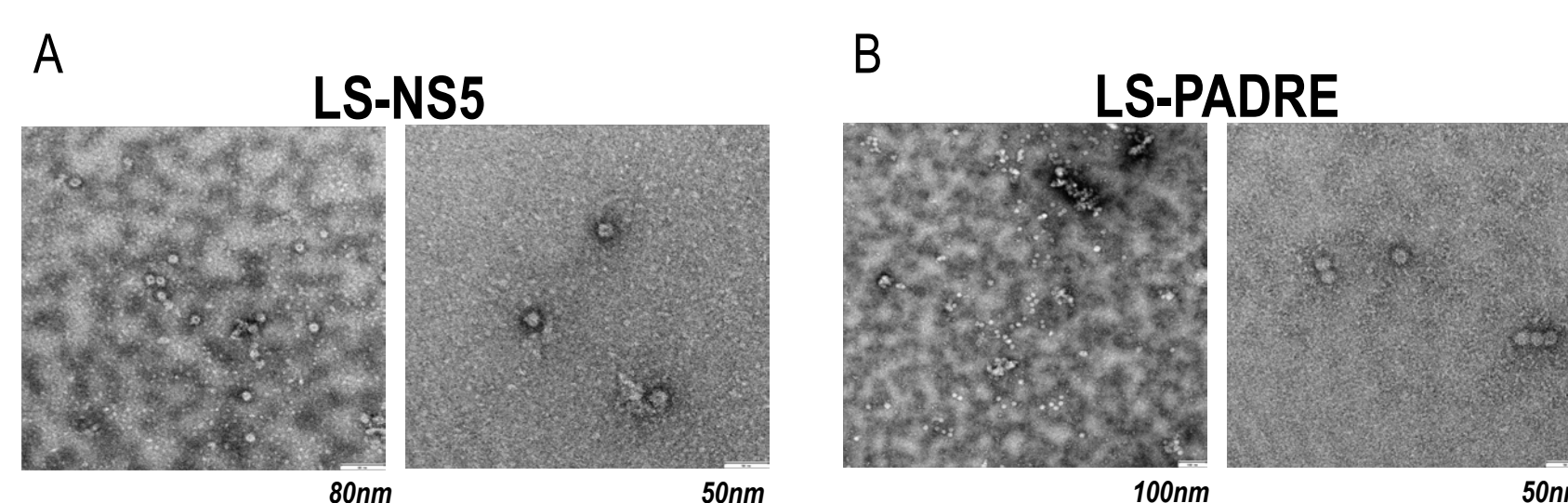


Figure 4. Morphology of LS-polypeptides. Imaged by transition electron microscopy (TEM). Constructs recombinantly expressed in *E.coli* and purified using affinity chromatography. Expected diameter: 20-30nm. (A) LS-NS5 nanoparticle (B) LS-PADRE nanoparticle

- T-cell polypeptides did not interfere with the stereochemistry of LS when using the SpyCatcher/SpyTag conjugation system (Fig 5).

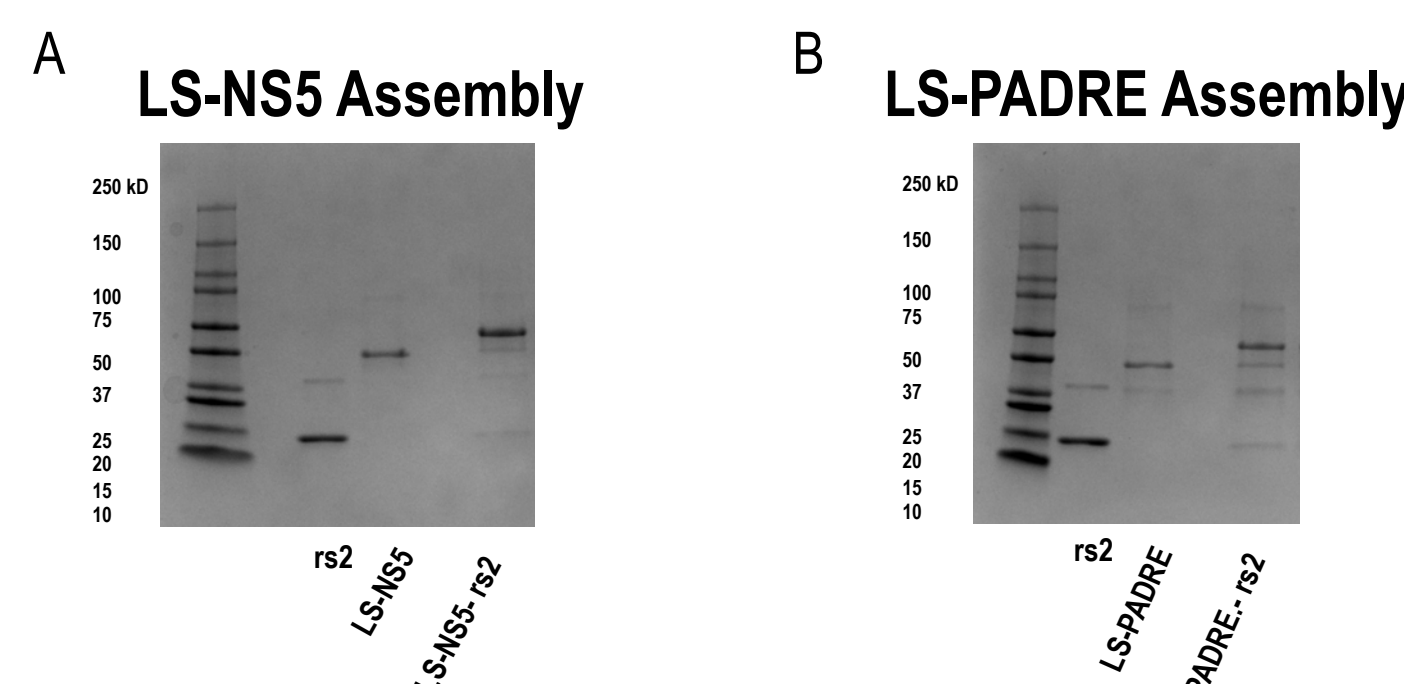


Figure 5. Assembly of LS-polypeptide and rs2 using SpyCatcher/SpyTag conjugation system. SDS PAGE gel stained with Coomassie Blue. (A) LS-NS5 assembled with free rs2 (B) LS-PADRE assembled with free rs2.

Results & Discussion cont'd

- Mice were immunized with combinations of LS-polypeptide, with and without conjugated rs2 (Fig 6).

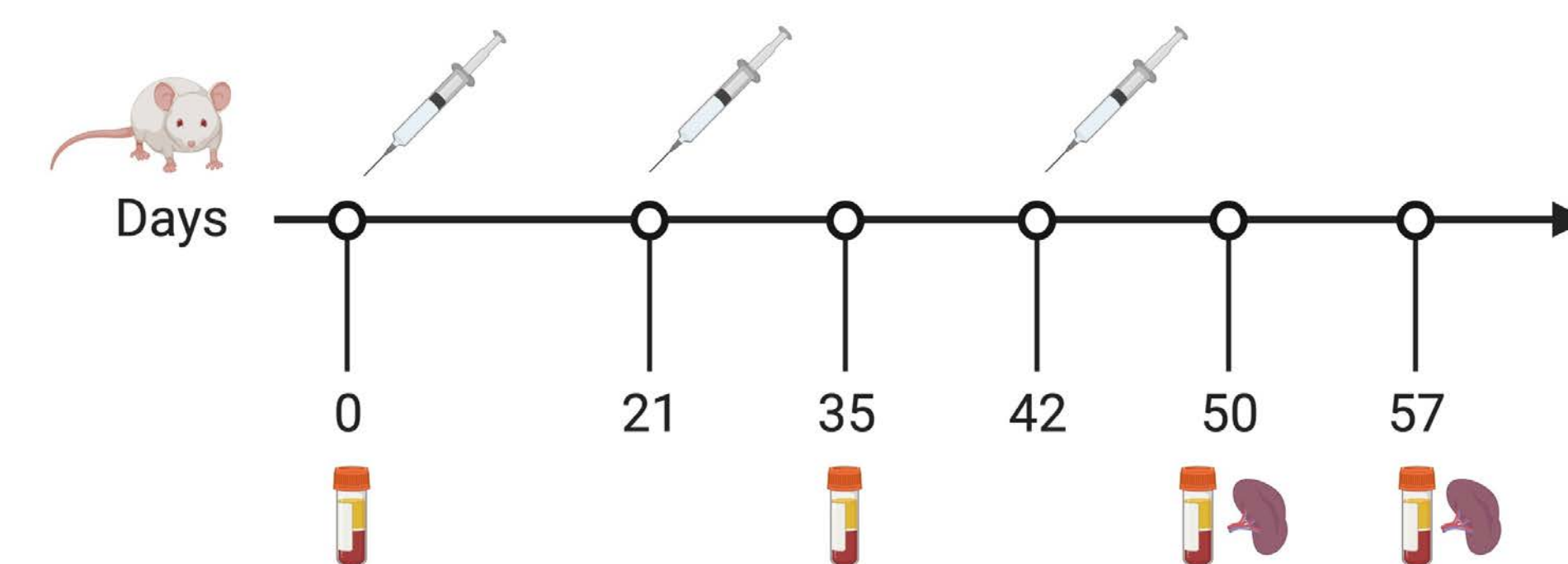


Figure 6. Immunization scheme ³. Five (5) groups; n=5; Female Balb/C mice. Immunogens were delivered by intraperitoneal (IP) injections for a total of three injections, each 21 days apart. 1 prime (14.5 ug) and 2 boosts (7.25 ug). Blood samples were collected on days 0 and 35 by submandibular bleeding. On day 50 and 57 blood samples were collected by cardiac punctures and spleens were removed for T-cell analysis.

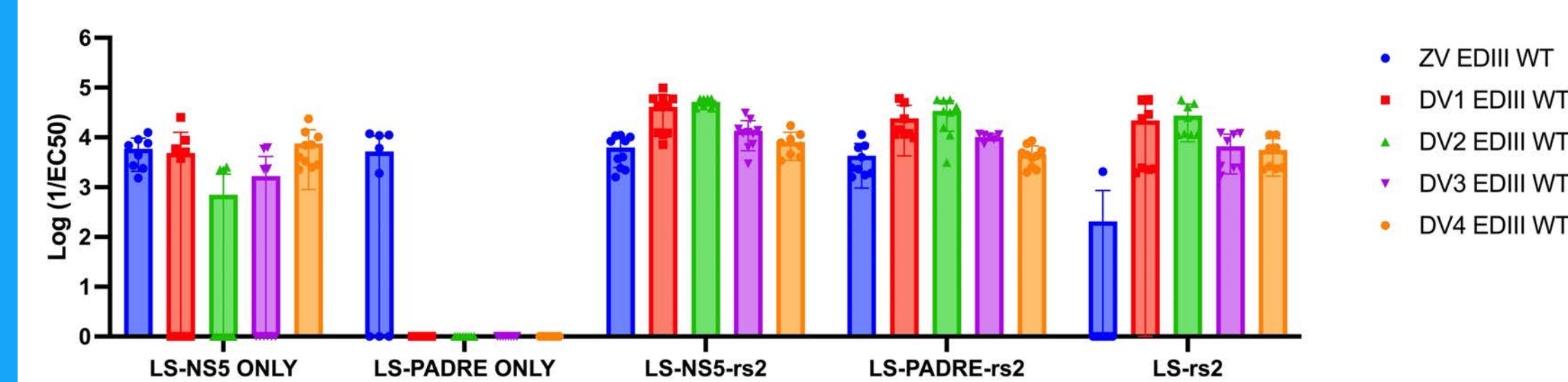


Figure 7. ELISA EC50 values of D35 sera. ELISAs were done per mouse, twice in technical duplicates. Reactivities against ZIKV & DENV 1-4 EDIIIs were evaluated.

- Antibody reactivity to rs2 was preserved with the presence of T-cell polypeptide immunogens.
- Sera from LS-PADRE ONLY lacked reactivity to DIII, while LS-NS5-ONLY showed variable reactivity to DIIIs (Fig 7).

Future Directions

- Characterize neutralizing properties of sera antibodies.
- Perform T-Cell Stimulation assays using phorbol myristate acetate (PMA) – ionomycin and free peptides.
- Further explore other LS - polypeptide constructs.
- Optimize the immunization scheme.

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